

Linear-Algebraic Analysis of a High-Dimensional Spectroscopic Dataset: Cleaning, Principal Component Analysis, and Clustering

SIMC 2026 Endeavour Challenge — Question 3, Parts (a)–(e)

May 2026

Abstract

We analyse a design matrix of $N = 1000$ infrared spectra, each recorded at $M = 3401$ frequencies. We first remove two classes of corrupted measurements by thresholding the distribution of row norms at its spectral gaps, retaining $\tilde{N} = 924$ clean spectra. After mean-centering we form the covariance matrix $C = \frac{1}{\tilde{N}} \tilde{X}^\top \tilde{X}$ and prove that its diagonal records the per-feature variances and that its trace, the total variance, is conserved under orthogonal change of basis. Diagonalising C (principal component analysis) yields a spectrum with a sharp elbow at $k^* = 8$ signal eigenvalues and a dominance ratio $\lambda_1/\lambda_2 \approx 3.53$; we show that the conventional “95% of variance” rule is misleading here, requiring 667 mostly-noise components. The leading eigenvector is a contrast direction supported on two narrow molecular bands and is shown to be a difference-spectrum, essentially orthogonal to the mean spectrum. Projecting the data onto the four leading eigenvectors resolves the samples into approximately 25 well-separated clusters, each a distinct chemical concoction.

1 Setup and notation

The data form a *design matrix* $X \in \mathbb{R}^{N \times M}$ whose rows are *samples* (one vial of a liquid sample) and whose columns are *features* (one infrared frequency). The entry X_{ij} is the absorbance of sample i at frequency j . We take $N = 1000$ and $M = 3401$.

Remark 1.1. The problem statement quotes $M = 3041$, whereas the supplied array `mystery_matrix1.npy` has a 3401-long feature axis; the two figures are a digit transposition. We trust the data and use $M = 3401$ throughout, identifying the sample axis as the one of length 1000. The file is stored transposed (shape 3401×1000), so we work with its transpose so that rows index samples.

Definition 1.1 (Euclidean norm). For $x = (x_1, \dots, x_M) \in \mathbb{R}^M$, the Euclidean norm is $\|x\| = (\sum_{j=1}^M x_j^2)^{1/2}$, the length of x . It is a single scalar summarising the overall magnitude of a spectrum.

2 Part (a): cleaning the data

Empty vials

A vial containing no sample yields only low-amplitude background noise: a nearly flat spectrum of small values and hence a small norm. A genuine spectrum has sharp absorption peaks and a large norm. Sorting the 1000 row norms exposes a clean gap: 40 measurements lie at $\|x\| \approx 1.6$, after which there is a void until the bulk begins at $\|x\| \approx 4.3$ (Figure 1, left). We set the threshold at the midpoint of this largest sub-median gap, a choice that requires no hand-tuned constant and is stable under perturbation. This isolates exactly $T = 40$ empty vials, which are precisely rows 960–999.

A second error: high-norm outliers

The upper tail of the norm distribution contains a second, clearly detached group of 36 measurements at $\|x\| \approx 8.9$, separated from the bulk by a gap roughly five times wider than any internal gap and occupying the contiguous block of rows 924–959. We flag these as a second error. We note that a coherent high-magnitude group could also be a genuinely strong concoction; we therefore (i) provide the independent evidence of Section 6 that they form a single isolated cluster whose removal lowers the cluster count by exactly one, and (ii) keep the removal reversible.

Selection by Boolean mask

Writing $\mathbf{1}[\cdot]$ for the indicator, we retain sample i iff $\tau_{\text{lo}} \leq \|x_i\| \leq \tau_{\text{hi}}$ with $\tau_{\text{lo}} \approx 2.98$, $\tau_{\text{hi}} \approx 8.15$. Indexing by this Boolean mask (rather than deleting rows) is non-destructive, composes the two criteria in one operation, and preserves original sample indices. The cleaned matrix has $\tilde{N} = 924$ rows.

Validation

The removed spectra are flat noise, the retained ones peaked. Quantitatively, the empty vials are 50.2% exact zeros against 24.5% for genuine spectra: background noise is symmetric about zero and the instrument clips negative absorbance to zero, so a zero-mean noise channel is exactly 0 with probability $\approx \frac{1}{2}$, a signature absent from peak-bearing data.

3 Part (b): variance, mean-centering, and the covariance matrix

Definition 3.1 (Mean-centering). With column means $\bar{X}_j = \frac{1}{\tilde{N}} \sum_{i=1}^{\tilde{N}} X_{ij}$, the centered matrix is $\tilde{X}_{ij} = X_{ij} - \bar{X}_j$, so every feature has zero mean.

Mean-centering is essential: without it the dominant direction of variation would be the mean spectrum shared by all samples, which carries no discriminative information. We study deviations from the mean, where the discriminating structure lives.

Definition 3.2 (Covariance matrix). $C = \frac{1}{\tilde{N}} \tilde{X}^\top \tilde{X} \in \mathbb{R}^{M \times M}$, with $C_{ij} = \frac{1}{\tilde{N}} \sum_{k=1}^{\tilde{N}} \tilde{X}_{ki} \tilde{X}_{kj}$.

Proposition 3.1. C is symmetric, and its diagonal entries are the per-feature variances: $C_{jj} = \text{Var}_j := \frac{1}{\tilde{N}} \sum_k \tilde{X}_{kj}^2$.

Proof. Symmetry: $C^\top = \frac{1}{\tilde{N}} (\tilde{X}^\top \tilde{X})^\top = \frac{1}{\tilde{N}} \tilde{X}^\top \tilde{X} = C$. Setting $i = j$ in the definition gives $C_{jj} = \frac{1}{\tilde{N}} \sum_k \tilde{X}_{kj}^2$, which, as the data are centered, is exactly the variance of feature j . $\square$

Proposition 3.2 (Conservation of total variance). *The total variance $\text{tr}(C) = \sum_j C_{jj}$ is invariant under any orthogonal change of basis $C \mapsto Q^\top C Q$ ($Q^\top Q = I$); in particular it equals the sum of the eigenvalues of C .*

Proof. $\text{tr}(Q^\top C Q) = \text{tr}(C Q Q^\top) = \text{tr}(C)$ by cyclicity of the trace and $Q Q^\top = I$. Diagonalising $C = Q \Lambda Q^\top$ gives $\text{tr}(C) = \text{tr}(\Lambda) = \sum_i \lambda_i$. $\square$

Empirically $\text{tr}(C) \approx 7.90$. The variance profile (Figure 2) is concentrated in a few narrow bands — the molecular absorption regions — with the global maximum at feature $j^* = 910$, where $C_{j^*j^*} \approx 0.0199$. Most features are nearly flat, so the data is effectively low-dimensional.

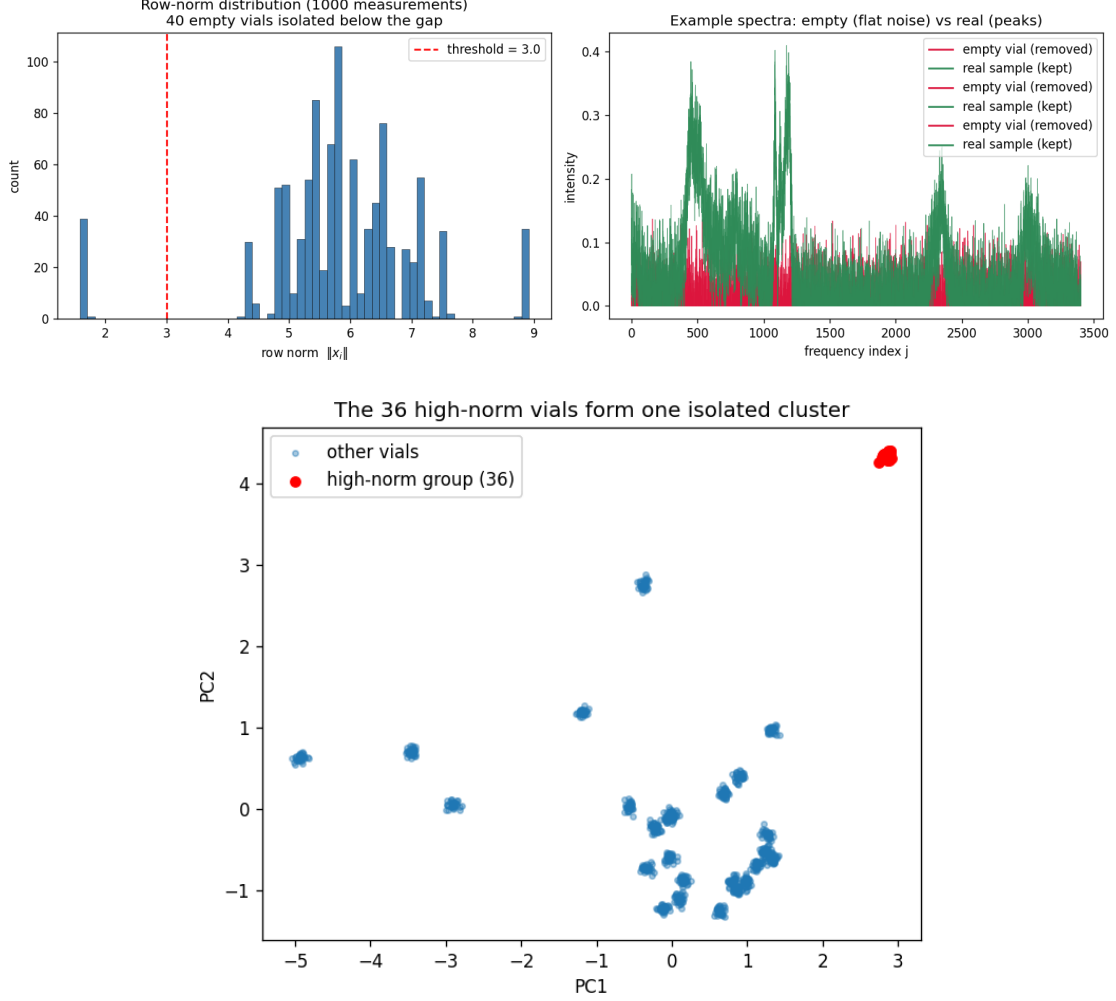


Figure 1: *Top*: row-norm distribution; 40 empty vials sit below the gap, with example empty (flat) and genuine (peaked) spectra. *Bottom*: the 36 high-norm vials form one tight, isolated cluster in the leading principal-component plane.

4 Part (c): the eigenspectrum (PCA)

Definition 4.1 (Eigenpair). (λ, ν) with $\nu \neq 0$ is an eigenpair of C if $C\nu = \lambda\nu$.

Proposition 4.1 (Eigenvalues are variances). *For a unit vector w , the variance of the data projected onto w is $w^\top Cw$. If $C\nu = \lambda\nu$ with $\|\nu\| = 1$ then $\nu^\top C\nu = \lambda$.*

Proof. The projection of sample $\tilde{x}_i$ onto w is $z_i = \tilde{x}_i^\top w$, with mean zero; its variance is $\frac{1}{N} \sum_i z_i^2 = \frac{1}{N} w^\top \tilde{X}^\top \tilde{X} w = w^\top Cw$. Substituting $C\nu = \lambda\nu$ and $\|\nu\| = 1$ gives $\nu^\top C\nu = \lambda\|\nu\|^2 = \lambda$. $\square$

Proposition 4.2. C is positive semi-definite; hence all $\lambda_i \geq 0$.

Proof. For any w , $w^\top Cw = \frac{1}{N} \|\tilde{X}w\|^2 \geq 0$. Applying this to a unit eigenvector and using Proposition 4.1 gives $\lambda = \nu^\top C\nu \geq 0$. $\square$

By the spectral theorem C (real symmetric) has real eigenvalues and an orthonormal eigenbasis. We compute the decomposition with `numpy.linalg.eigh`, which is specialised for symmetric matrices — guaranteeing real eigenvalues and orthonormal eigenvectors, and faster and more stable than `eig` — then sort the eigenvalues in decreasing order and clip rounding-level negatives to zero (justified by Proposition 4.2).

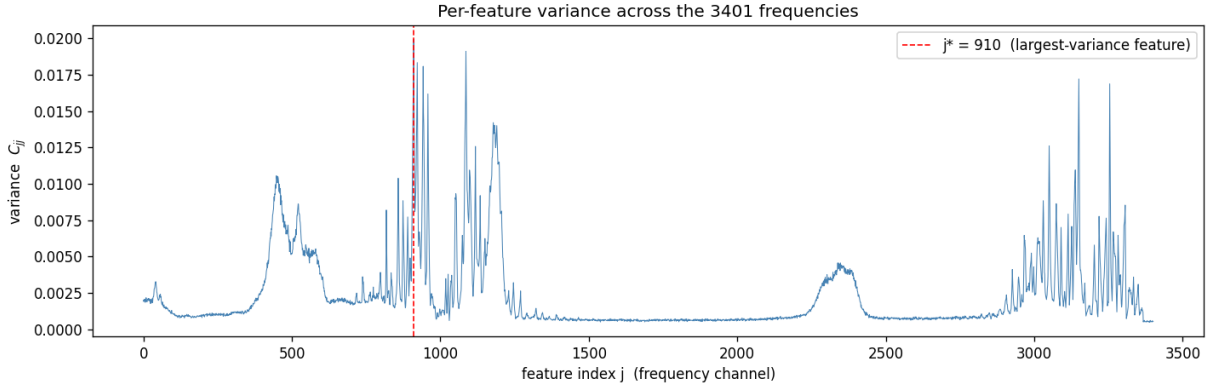


Figure 2: Per-feature variance C_{jj} . Variance concentrates in narrow molecular bands; the red line marks the maximal-variance feature $j^* = 910$.

k	1	2	3	4	5	6	7	8	9
λ_k	2.492	0.705	0.590	0.254	0.123	0.105	0.078	0.039	0.0093

Table 1: Leading eigenvalues. The drop $\lambda_8/\lambda_9 \approx 4.1$ is the largest in the spectrum and marks the signal/noise boundary.

Dominance ratio and the elbow

The dominance ratio is $\lambda_1/\lambda_2 \approx 3.53$: one direction captures far more variance than any other, indicating strong low-dimensional structure (and a wide spectral gap, the condition for rapid convergence of the Power Method of Question 2). We locate the *elbow* as the largest multiplicative drop between consecutive eigenvalues — a ratio, since the signal/noise transition is multiplicative on the logarithmic scale of Figure 3. The maximal drop is $\lambda_8/\lambda_9 \approx 4.1$ (Table 1), giving $k^* = 8$: λ_8 remains $\approx 4\times$ above the noise floor (signal), while λ_9 already lies on the floor (noise). This agrees with the ~ 10 dominant molecules: a few chemical components span only a few independent directions.

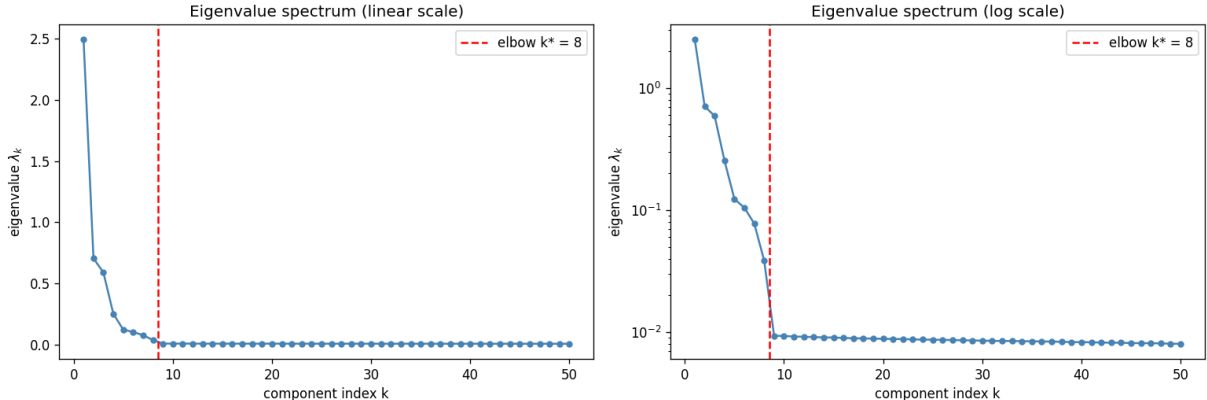


Figure 3: Eigenvalue spectrum on linear (left) and logarithmic (right) scales. The logarithmic plot exhibits a sharp elbow at $k^* = 8$, followed by a flat noise floor.

Cumulative variance: a caution

Let the explained-variance ratio be $\lambda_k/\sum_i \lambda_i$. The eight signal components account for only $\approx 55.5\%$ of the total variance; reaching 95% requires $k = 667$ components (Table 2, Figure 4).

The reason is that the small noise floor is spread over ≈ 900 dimensions (the rank is at most $\tilde{N}-1$), so its tiny per-component contributions accumulate. The naive “retain 95% of variance” heuristic therefore grossly overestimates the dimensionality; the elbow $k^* = 8$ is the honest measure. As a check, $\sum_i \lambda_i \approx \text{tr}(C) \approx 7.90$ (Proposition 3.2).

cumulative variance	50%	80%	90%	95%	99%
components k	4	309	516	667	850

Table 2: Components required to reach each variance threshold; contrast the 8 signal components (55.5%) with the 667 needed for 95%.

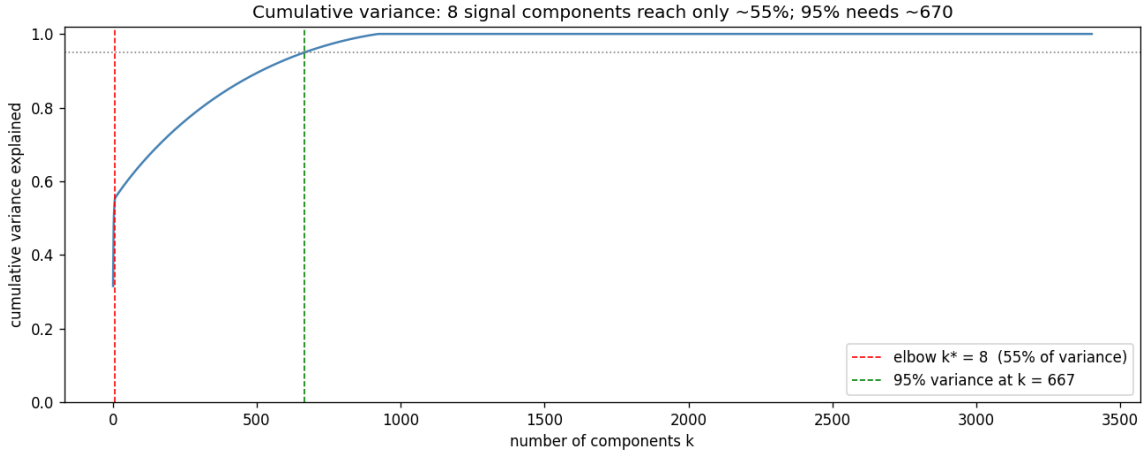


Figure 4: Cumulative variance versus k . The elbow (red) reaches only $\approx 55\%$; 95% (green) is not attained until $k = 667$.

5 Part (d): the leading eigenvector

The leading eigenvector $\nu_1 \in \mathbb{R}^M$ lies in feature space and assigns a *loading* to each frequency; equivalently it is the direction of greatest variance. The associated *scores* $z_i = \tilde{x}_i \cdot \nu_1$ are, by Proposition 4.1, the maximally-spread one-dimensional summaries of the samples. Eigenvectors are defined only up to sign ($C(-\nu) = \lambda(-\nu)$); we fix the sign so the largest-magnitude loading is positive.

Figure 5 (left) shows that ν_1 is supported on two narrow bands ($j \approx 900-960$ and $j \approx 3150-3260$) and takes both signs ($\approx 27\%$ of loadings negative). Mixed signs make ν_1 a *contrast*: it measures the balance of absorption between two groups of frequencies rather than overall intensity.

Definition 5.1 (Cosine similarity). $\cos(a, b) = \frac{a \cdot b}{\|a\| \|b\|} \in [-1, 1]$, the cosine of the angle between a and b ; it is scale-invariant and compares direction, not magnitude.

We find $\cos(\nu_1, \tilde{x}_{193}) \approx 0.92$ for the best-matching *centered* spectrum, but only ≈ 0.67 against the corresponding *raw* spectrum and ≈ 0.03 against the mean spectrum. Because ν_1 lives in the centered space it is essentially orthogonal to the (uncentered) mean, which dominates raw spectra; hence ν_1 resembles a *difference-spectrum* — the way the most-deviated concoction departs from the mean (Figure 5, right). This mirrors Question 1, where the dominant eigenvector was the long-time steady state of the dynamics.

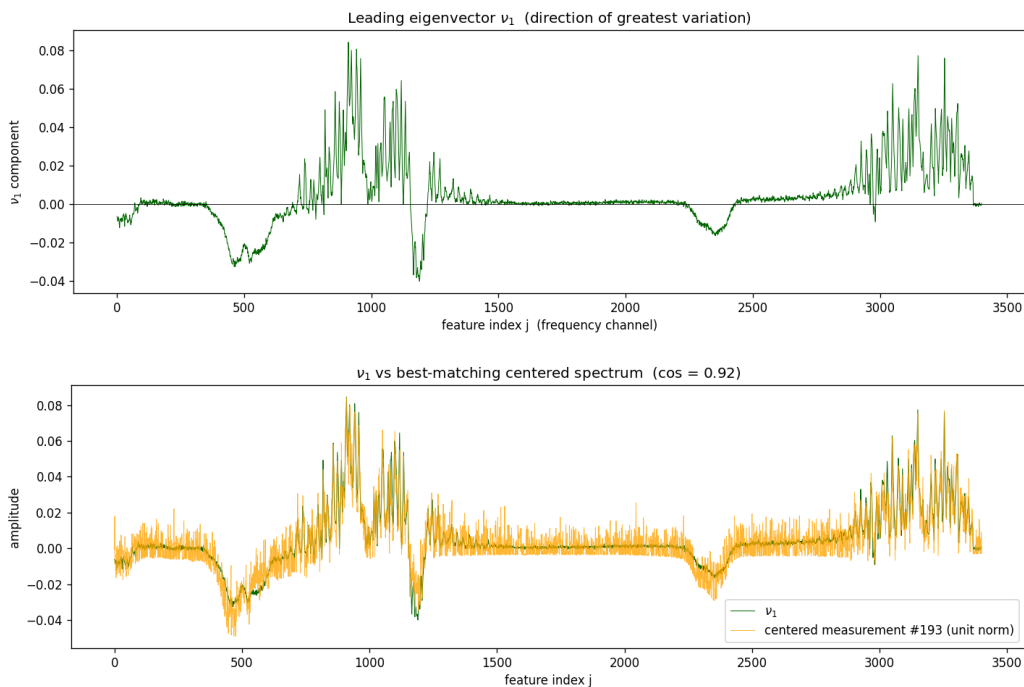


Figure 5: *Top*: loadings of ν_1 , concentrated in two molecular bands with mixed signs (a contrast direction). *Bottom*: ν_1 overlaid with the best-matching centered spectrum ($\cos \approx 0.92$).

6 Part (e): clustering in covariance space

We project onto the four leading eigenvectors. With $V_4 = [\nu_1 \nu_2 \nu_3 \nu_4] \in \mathbb{R}^{M \times 4}$, the compressed coordinates are $Z = \tilde{X}V_4 \in \mathbb{R}^{\tilde{N} \times 4}$; row z_i holds sample i ’s four scores. This is dimensionality reduction from 3401 to 4 retaining the dominant variation. The $\binom{4}{2} = 6$ pairwise scatter plots (Figure 6) reveal many tight, well-separated groups.

To count them without external libraries we use leader clustering: traversing the points, a new cluster centre is opened whenever a point lies farther than a radius r from all existing centres. With r set to a few multiples of the median nearest-neighbour spacing the count is stable at 25 for all $r \in [6, 12] \times$, so it reflects the data rather than the threshold. Hence the 924 cleaned vials comprise ≈ 25 distinct concoctions — comfortably below the “fewer than 100 unique samples” of Part (h). Although $\nu_1, \dots, \nu_4$ live in feature space, projecting samples onto them exposes structure in sample space, because chemically similar spectra receive similar scores. Part (f) repeats the construction with the Gram matrix $G = \frac{1}{\tilde{N}} \tilde{X} \tilde{X}^\top$, which shares the nonzero eigenvalues of C (Question 2d) and should recover the same groups.

7 Summary

Cleaning by norm-gap thresholding removed 40 empty vials and 36 high-norm outliers, leaving $\tilde{N} = 924$ spectra. The covariance matrix has total variance $\text{tr}(C) \approx 7.90$, a dominance ratio $\lambda_1/\lambda_2 \approx 3.53$, and a signal subspace of dimension $k^* = 8$; the 95%-variance heuristic is shown to be inappropriate. The leading eigenvector is an interpretable difference-spectrum, and projection onto the top four eigenvectors resolves the data into ≈ 25 chemical concoctions. All computations are reproduced in the accompanying notebook `SIMC_2026.ipynb`.

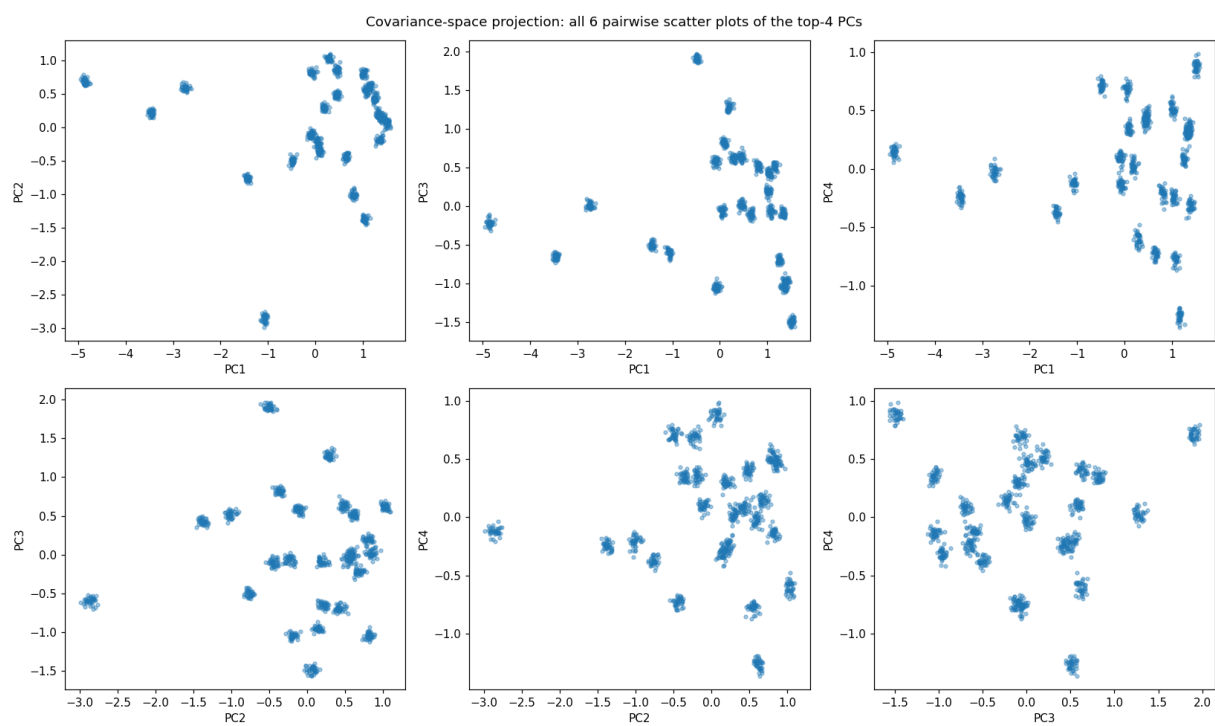


Figure 6: The six pairwise scatter plots of the top-4 principal-component scores. Each point is one vial; the data resolves into ≈ 25 well-separated clusters.